

On 180M voxels the residual is one edge-pair selection swap

Stage / surface	What is counted	Residual	Of	Reading
Energy scale-index	mismatched voxels	0	16.8M	exact discrete field
Energy float	max $ \Delta $	$\sim 2e-11$	allclose	library FP drift only
Vertices	position / scale	0	exact	bit-identical geometry
Edges ownership	disagreed voxels	~ 8	5.84M claimed	99.9999% map agree
Edges connections	pair multiset Δ	1	69,500	one equal-metric swap
Network strands	strand multiset Δ	1	48,049	downstream of that swap

Takeaway: Energy/Vertices are closed. Edges ownership far exceeds the $\geq 60\%$ ADR 0012 bar. The only open multiset residual is one degree-pruning pair swap on a shared hub vertex (crop: MATLAB [4212, 6281] vs Python [4043, 6281]) — Network's -1 strand is the same event.